

COORDINATION-DEPENDENT HYDROGEN ADSORPTION ON Co-, Fe-, AND Ni-FUNCTIONALIZED MONOLAYER CrCl_3

Spin-Polarized DFT, Coordination Trade-Offs, and Sabatier Electrocatalysis

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Focus: HER Electrocatalysis & Spin-Coordination Engineering



Motivation: 2D Magnetic Catalysts for Green H₂

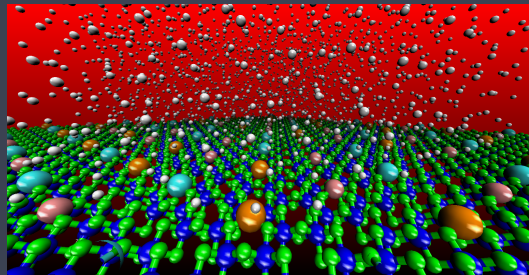
Acidic Hydrogen Evolution Reaction (HER)

Cathodic reaction: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$, $\eta_{\text{HER}} = |\Delta G_{\text{H}^*}|/e$

- ⊙ **Sabatier Ideal:** Near-thermoneutral binding ($\Delta G_{\text{H}^*} \approx 0$).
- ⊙ **Benchmark Pt (111):** $\Delta G_{\text{H}^*} = -0.090$ eV (scarce, costly).

Why Monolayer CrCl₃?

- ⊙ **2D Magnetic Semiconductor:** Exfoliable vdW crystal with intrinsic FM order ($\mu_{\text{Cr}} \approx 3 \mu_{\text{B}}$) offering a spin degree of freedom.
- ⊙ **Active-Site Dilemma:** Saturated Cl basal plane is chemically inert $\Rightarrow$ Single TM atom engineering creates localized active centers!



TOC Concept: Single-atom functionalization & coordination engineering.

Spin-Polarized DFT & AIMD Dynamics

- ⊙ **Code & PAW:** VASP, PAW, GGA-PBE.
- ⊙ **Supercell:** $2 \times 2 \times 1$ ($\text{Cr}_8\text{Cl}_{24}$), 15 Å vacuum, $E_{\text{cut}} = 400$ eV, $5 \times 5 \times 1$ Γ-mesh, residual forces < 0.025 eV/Å.
- ⊙ **Hubbard U Benchmark:** Dudarev scheme ($U^* = 3.29$ eV) reproduces experimental optical gap (2.56 eV) and linear response ($U_{\text{LR}} = 3.32$ eV) + Grimme DFT-D3(BJ).
- ⊙ **AIMD 300 K:** 5 ps NVT (Nosé-Hoover, 1 fs step) for finite-temperature kinetic trapping and structural stability.

Computational Hydrogen Electrode (CHE)

$$\Delta G_{\text{H}^*} = E_{\text{surf+H}} - E_{\text{surf}} - \frac{1}{2}E_{\text{H}_2} + (\Delta E_{\text{ZPE}} - T\Delta S)$$

Full phononic thermochemistry via explicit Hessians (IBRION=5).

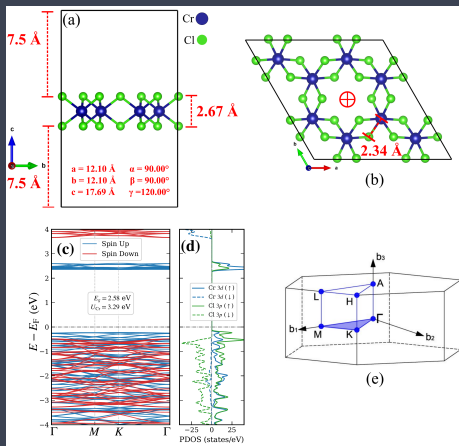
Distortion-Interaction Decomposition

Decouples intrinsic chemical binding from substrate distortion:

$$\Delta E_{\text{ads}}^{\text{H}} = E_{\text{int}}^{\text{H}} + E_{\text{def}}$$

- ⊙ $E_{\text{def}} = E_{\text{surf}}^{\text{def}} - E_{\text{surf}} \geq 0$ (lattice distortion strain).
- ⊙ $E_{\text{int}}^{\text{H}} = E_{\text{surf+H}} - E_{\text{surf}}^{\text{def}} - E_{\text{H}} < 0$ (intrinsic affinity).

Pristine Monolayer CrCl₃: Structure & Catalytic Inertness



Crystallographic Architecture

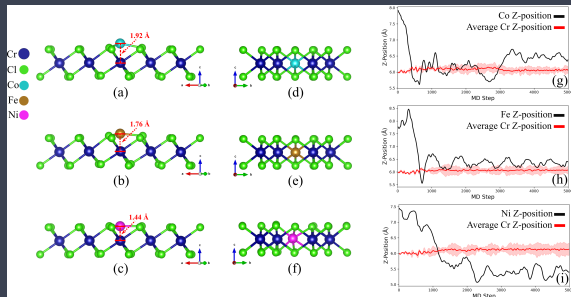
- ⊙ Honeycomb network: $a = 12.10 \text{ Å}$ (2×2), thickness $h = 2.67 \text{ Å}$, edge-sharing CrCl_6 ($d = 2.34 \text{ Å}$).
- ⊙ FM ground state: $M_{\text{tot}} = 24.0 \mu_B$ ($3 \mu_B/\text{f.u.}$, $\mu_{\text{Cr}} \approx 2.94 \mu_B$), direct gap $E_g = 1.50 \rightarrow 2.56 \text{ eV}$ (PBE+U).

Basal Plane Catalytic Inertness

- ⊙ S_1 (Top-Cl): Spontaneous H repulsion ($d_{\text{H-surf}} = 3.42 \text{ Å}$).
- ⊙ S_2 (Hollow): Geometrically unstable site.
- ⊙ S_3 (Top-Cr): Metastable contact, but severely endergonic: $\Delta G_{\text{H}^*} = +2.44 \text{ eV}$ ($\eta = 2.44 \text{ V}$).
- ⊙ Pristine basal plane is completely inert for HER!

Coordination Motifs: Surface-Adsorbed vs. Interstitial Embedded

Two Distinct Local Coordination Motifs

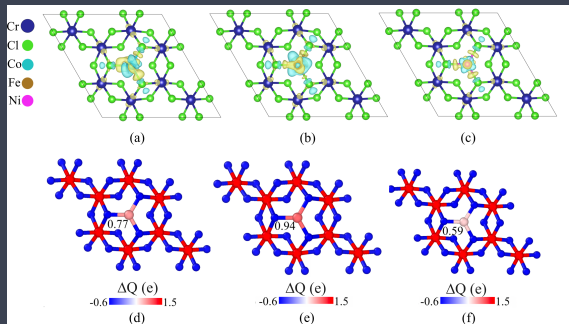


- ⊙ **Surface-Adsorbed (3-fold Hollow):** TM exposed ($z = 1.44\text{--}1.92 \text{ \AA}$). $E_{\text{bind}} = -2.82 \text{ (Co)}, -2.60 \text{ (Fe)}, -3.38 \text{ eV (Ni)}$.
- ⊙ **Embedded (6-fold Pore):** Core penetration ($z \approx 0$). $E_{\text{bind}} = -3.55 \text{ (Co)}, -3.36 \text{ (Fe)}, -4.95 \text{ eV (Ni)}$. Thermodynamic sink ($\Delta E_{\text{emb-ads}} < 0$).

Room-Temperature Kinetic Trapping (AIMD 300 K)

- ⊙ **Co & Fe:** Metastable at surface ($z > 0$), protected by diffusion barriers.
- ⊙ **Ni:** Spontaneously penetrates within 1 ps into core pore ($z \rightarrow 0 \text{ \AA}$).

Charge Redistribution & In-Gap Electronic States



Net Charge Transfer & Spin Polarization

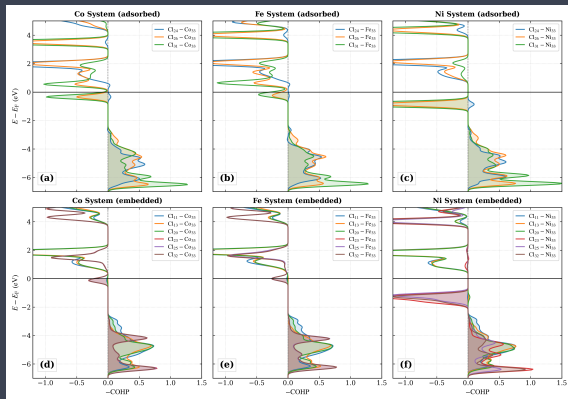
$\Delta\rho = \rho_{\text{tot}} - \rho_{\text{host}} - \rho_{\text{TM}}$ (electron donors to CrCl_3 host):

- ⊙ **Fe (ads):** $\Delta Q = +0.94 e$, $\mu = +3.34 \mu_B$ (parallel to Cr).
- ⊙ **Co (ads):** $\Delta Q = +0.77 e$, $\mu = +1.99 \mu_B$ (parallel to Cr).
- ⊙ **Ni (ads):** $\Delta Q = +0.59 e$, $\mu = -0.76 \mu_B$ (antiparallel to Cr).

In-Gap Frontier States

- ⊙ 3-fold surface adatoms induce sharp, spin-polarized TM-3d states at E_F , activating the surface for proton reception.
- ⊙ 6-fold pore embedding shifts 3d states deep into the valence band ($-3 \dots -1$ eV), depopulating E_F .

Chemical Bonding: -COHP and the Coordination Trade-Off



Crystal Orbital Hamilton Population (-COHP)

$$\text{ICOHP}_{\text{TM-Cl}} = \int_{-\infty}^{E_F} \text{COHP}(E) dE \text{ (covalent TM-3d/Cl-3p):}$$

- ⊙ Bonding manifold spans -6.8 to -2.5 eV below E_F .
- ⊙ **Surface (3-fold):** Strong individual bonds ($\text{ICOHP}_{\text{avg}} = -2.23$ eV for Co, -2.28 eV for Fe).
- ⊙ **Embedded (6-fold):** Weaker individual bonds ($\text{ICOHP}_{\text{avg}} = -1.87$ eV), but higher cumulative bonding ($\text{ICOHP}_{\text{cum}} \approx -11.4$ eV).

The Coordination Trade-Off Principle

High coordination stability does not imply catalysis!
Deep 6-fold coordination maximizes host integration at the cost of chemical passivity toward reactants.

Hydrogen Adsorption Thermodynamics: Sabatier Optimum

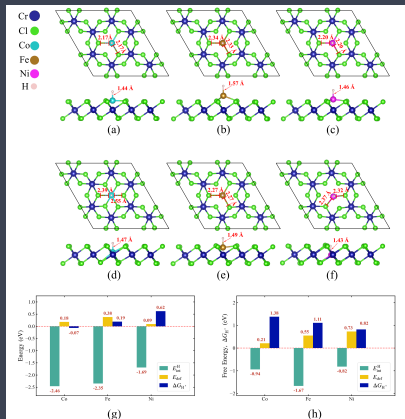
Sabatier Optimum: Near-Thermoneutral

PBE+D3+*U* thermochemistry with explicit phononics:

- ⊙ **Co (ads):** $\Delta G_{H^*} = -0.069$ eV ($\eta = 0.069$ V) — **outperforms Pt (111)** (−0.090 eV)!
- ⊙ **Fe (ads):** $\Delta G_{H^*} = +0.185$ eV ($\eta = 0.185$ V).
- ⊙ **Ni (ads):** $\Delta G_{H^*} = +0.620$ eV.

Embedded Motifs & Distortion Penalty

- ⊙ Embedding suppresses activity: $\Delta G_{H^*} \geq +0.82$ eV.
- ⊙ Co binds strongly ($E_{int}^H = -2.46$ eV) with low strain ($E_{def} = 0.18$ eV).
- ⊙ Fe binds well (−2.35 eV) but incurs double distortion ($E_{def} = 0.38$ eV).



Descriptor Analysis: *d*-Band Center & Spin Coupling

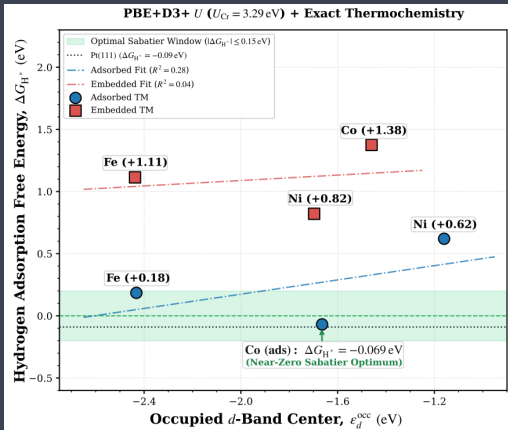
Occupied *d*-Band Center (ϵ_d^{occ})

$$\epsilon_d^{\text{occ}} = \frac{\sum_{\sigma} \int_{-\infty}^{E_F} (E - E_F) \rho_{d,\sigma}(E) dE}{\sum_{\sigma} \int_{-\infty}^{E_F} \rho_{d,\sigma}(E) dE}$$

- ⊙ **Surface Breakdown:** Ni has ϵ_d^{occ} closest to E_F (−1.15 eV), yet worst activity ($\Delta G = +0.62$ eV).
- ⊙ **Embedded Breakdown:** Fe has deepest ϵ_d^{occ} (−2.43 eV), yet lowest embedded ΔG_{H^*} (+1.11 eV).

Why Scaling Fails: Spin & Coordination

- ⊙ Exchange splitting & local coordination override single-band centroids ($R^2 \leq 0.28$).
- ⊙ **Spin Quenching:** Chemisorption quenches Co spin ($\Delta\mu = -0.98 \mu_B$), proving direct spin-channel coupling!



Benchmark & Catalyst Design Principles

System / Site	Coord.	E_{bind} (eV)		ΔG_{H^*} (eV)		η_{HER} (V)	μ_{TM} (μ_{B})	ΔQ_{Bader} (e)
		PBE	PBE+D3+U	PBE	PBE+D3+U			
Pristine S_1 (Top-Cl)	1-fold Cl	—	—	+1.79	+1.12	1.12	—	—
Pristine S_2 (Hollow)	3-fold Cl	—	—	+2.68	+2.04	2.04	—	—
Pristine S_3 (Top-Cr)	1-fold Cr	—	—	+1.88	+2.44	2.44	—	—
CrCl₃-Co (ads)	3-fold Cl	−2.82	−5.14	+0.18	−0.069	0.069	+1.99	+0.77
CrCl₃-Fe (ads)	3-fold Cl	−2.60	−6.13	+0.32	+0.185	0.185	+3.34	+0.94
CrCl ₃ -Ni (ads)	3-fold Cl	−3.38	−4.37	+0.81	+0.620	0.620	−0.76	+0.59
CrCl ₃ -Co (emb)	6-fold Cl	−3.55	−5.38	+1.76	+1.375	1.375	+2.09	+0.82
CrCl ₃ -Fe (emb)	6-fold Cl	−3.36	−7.53	+1.35	+1.114	1.114	+3.34	+1.02
CrCl ₃ -Ni (emb)	6-fold Cl	−4.95	−3.91	+1.85	+0.821	0.821	−1.02	+0.68
Benchmark Pt (111)	Terrace	—	—	−0.09	−0.09	0.090	—	—

Key Takeaways

- ⊙ **Coordination Trade-Off:** 3-fold surface exposure beats 6-fold embedding.
- ⊙ **Sabatier Peak:** Surface Co ($\Delta G_{\text{H}^*} = -0.069$ eV) outdoes Pt (111).

Design Guidelines

- ⊙ **Kinetic Trapping:** Ambient 300 K barriers preserve active surface states.
- ⊙ **Support Coupling:** Conductive carbon supports bypass host gap.

Key References & Institutional Acknowledgments

Primary Literature References

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Computing Facilities

UFABC Multiuser Computing Center (CCM) and
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Summary & Outlook: Single-Atom Catalysis on 2D Magnets

Sabatier Optimum

- ⊙ **Co (ads) Optimum:**

$$\Delta G_{\text{H}^*} = -0.069 \text{ eV}$$

- ⊙ Overpotential $\eta = 69 \text{ mV}$, outperforming benchmark Pt (111) ($\eta = 90 \text{ mV}$).
- ⊙ Minimal deformation penalty ($E_{\text{def}} = 0.18 \text{ eV}$).

Coordination Trade-Off

- ⊙ **3-fold Surface:** Preserves active frontier $3d$ states at E_{F} .
- ⊙ **6-fold Core Pore:** Over-coordination causes electronic saturation ($\Delta G_{\text{H}^*} \geq +0.82 \text{ eV}$).
- ⊙ High bonding stability does **not** guarantee catalysis.

Spin-Coupled HER

- ⊙ **Moment Quenching:** Proton chemisorption systematically quenches TM spin ($\Delta\mu \approx -1.0 \mu_{\text{B}}$).
- ⊙ Direct spin-channel participation in electron transfer.
- ⊙ 2D magnets provide magnetic tunability for green H_2 .

Discussion & Contact

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